

Important:

1. Write a PL/SQL procedures to illustrate the CRUD operations with the transaction control like commit, rollback in the database using stored procedure.
2. Create user-defined PL/SQL functions of your choice for the given database and demonstrate their use in SQL queries and PL/SQL blocks.
3. Write a PL/SQL program to create INSERT, UPDATE, and DELETE triggers for the given database.
4. Implement indexes and sequences for the given database and demonstrate their use in improving query performance and generating unique key values.
5. Object and Object-Relational Databases (Oracle DB): - Create object types using Oracle Object Types. - Create object tables based on user-defined object types. - Insert, update and delete objects from object tables. - Create nested object attributes. - Create collection data types (VARRAY and Nested Table). - Implement object references (REF). - Create object methods (Member Functions and Procedures). - Query object tables using dot notation. - Implement inheritance using UNDER clause. - Write Object SQL queries for object retrieval.
6. Query Processing (Oracle DB): - Translate SQL queries into Relational Algebra. - Compare execution plans using EXPLAIN PLAN. - Perform different JOIN operations. - Implement aggregation queries. - Compare GROUP BY with HAVING. - Compare Nested Loop Join and Hash Join. - Analyze ORDER BY using large datasets. - Simulate pipelined query execution. - Execute queries with indexes and without indexes. - Compare query execution cost using AUTOTRACE.